Chapter 3

Financial Markets, Institutions, and Instruments

Financial markets are the institutions through which capital is allocated, risk is transferred, and prices are discovered. They allow households to save, firms to finance investment, governments to fund spending, and investors to diversify exposure. Anyone who wants to use data science or artificial intelligence in finance needs to understand these markets, because the objects of analysis are not abstract numbers but real contracts, incentives, and trading mechanisms.

Modern finance rests on a simple idea: people do not all have the same preferences, opportunities, or time horizons. Some want to consume today, others to save for the future, and still others will accept risk in exchange for compensation. Financial markets connect these needs. Without them, savings would be less productive, firms would find it harder to grow, and risk would be harder to manage.

This chapter introduces the structure of financial markets and the major classes of financial instruments. It begins with institutions and participants, moves to the securities traded in modern markets, and then explains how prices are formed in real trading systems, laying a foundation for the later chapters on pricing, portfolio management, risk, and trading.

Three worked examples anchor the more mechanical sections: a bid-ask spread decomposed into adverse-selection and inventory-cost components, a hand-computed walk through a small limit order book showing how an incoming order fills at one price or several, and a no-arbitrage bond-pricing example showing what happens when two ways of pricing the same cash flow disagree. Two case vignettes close the chapter: the 2010 Flash Crash, an afternoon when the mechanics of order flow and liquidity broke down in public, and the 2021 collapse of Archegos Capital Management, which reruns the chapter's margin arithmetic at a leverage of ten to one.

The institutional material (who trades, what they trade, and where) builds the vocabulary later chapters assume. Chapters 5 and 12 price and hold the instruments cataloged here; Chapters 13 and 14 return to the limit order book of Section 3.2.2 at execution and then microsecond speed; and the no-arbitrage logic of Section 3.4.3 recurs in every valuation and pricing chapter that follows.

Readers comfortable with market structure may skim the descriptive sections, but the worked examples are used again later and are worth working through by hand once.

3.1 The Market and Its Participants

A market is an institution before it is a dataset. This part sets out what financial markets are for, who shows up in them, the firms that intermediate between those participants, and the contracts they actually trade.

3.1.1 Why Financial Markets Exist

A financial market exists because economic agents have different needs and time horizons. A household may wish to save for retirement, a firm may need funding for expansion, and a government may need resources for public projects. Markets match the supply of and demand for capital.

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They do more than move money from one person to another. Markets also let people shift risk across time and across states of the world. A farmer who fears a decline in crop prices can use a futures contract to lock in a sale price. A pension fund that wants stable long-term income can purchase government bonds. A young entrepreneur who needs capital to build a company can issue equity to investors who are willing to share the upside and downside of the venture.

Financial markets also provide information. Prices reflect current supply and demand, but they also encode expectations about future cash flows, interest rates, inflation, and corporate performance. Markets are therefore systems for aggregating dispersed information into prices that guide decisions, not merely venues for exchange.

A market is more useful when participants can trade quickly and with confidence, so liquidity, transparency, and regulation matter. If a market is illiquid, prices may be unstable and trading costs may be high. If it lacks transparency, participants may face information asymmetry and make poor decisions. If it lacks rules, trust can erode and market participants may withdraw from the system.

3.1.2 Who Participates in Financial Markets

Financial market participants have different objectives and constraints. Households and retail investors save for retirement, education, or other future needs. Firms issue securities to raise funding for investment. Governments borrow to finance public spending or manage budget imbalances. Financial institutions such as banks, mutual funds, pension funds, and insurers intermediate between those who provide capital and those who need it. Chapter 1's Table 1.5 previewed these participant roles at a glance; this section develops each one in more depth.

Market makers play a particularly important role. A market maker provides liquidity by standing ready to buy and sell securities, usually quoting both a bid price and an ask price. In return for bearing inventory risk and exposure to adverse price movements, market makers earn the spread between the two prices, which is both a trading cost and compensation for providing immediacy and continuity.

Arbitrageurs and hedge funds also influence market behavior. Arbitrageurs exploit pricing discrepancies across related assets or markets. Hedge funds often pursue more complex strategies involving leverage, short selling, or macroeconomic views. Their activity can improve pricing efficiency but can also increase volatility when markets are under stress.

Finally, regulators and central banks shape the environment. Regulators enforce rules on disclosure, conduct, and market integrity. Central banks influence short-term interest rates and can intervene in crises to support liquidity or stabilize the financial system. Market outcomes therefore depend on public policy as well as private incentives.

3.1.3 Major Financial Institutions

Financial markets are organized around specialized institutions. Banks provide lending services, facilitate payments, and transform maturity and liquidity profiles. Asset managers collect capital from investors and allocate it across portfolios of securities. Insurance companies pool risk by charging premiums in exchange for compensation when specified losses occur. Exchanges provide venues where orders can be matched, and central clearinghouses reduce counterparty risk by interposing themselves between buyers and sellers. Chapter 1's Table 1.3 summarized these institutions' roles at a glance; this section develops each one further.

These institutions have different business models and incentives. A commercial bank earns money by borrowing at one rate and lending at another. An asset manager earns fees by managing assets for clients. An insurance company earns premiums and invests the proceeds while managing claims. Each therefore faces its own mix of credit, market, liquidity, and operational risk.

The financial system is fragile when institutions become highly interconnected. A bank may lend to a hedge fund, which may trade derivatives with a broker, which may depend on short-term funding from another bank. Such links make the system more efficient in normal conditions but transmit shocks quickly under stress, which is one reason understanding institutions is essential to understanding financial crises.

These institutions also shape the real economy. When banks tighten credit, firms may reduce investment. When insurers become more cautious, some risks become harder to insure. When exchanges improve market structure, trading costs may fall and liquidity improve. Such changes affect economic activity, corporate decisions, and household welfare.

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One institutional shift of the past fifteen years deserves its own mention, because it has moved a growing share of lending outside every venue described so far. Since the 2008 crisis, tighter bank capital regulation has pushed banks out of much corporate lending. Dedicated *private credit* funds, asset managers raising long-lockup capital to lend directly to mid-sized companies, have grown into a multi-trillion-dollar market filling that space, alongside the older private equity industry that buys entire companies rather than lending to them. These private markets differ from everything else in this chapter in a way that matters greatly for this book: they generate almost none of the data its methods assume. A private loan trades rarely or never, so there is no price history. The borrower files no public statements, so there is nothing for ratio analysis beyond what the lender privately negotiates access to. And a fund holds tens of positions, not the thousands of observations a statistical credit model trains on. AI's footprint in private markets is accordingly real but narrow: mostly natural language processing applied to deal documents and covenants, and screening tools for sourcing borrowers, while underwriting and valuation remain judgment-heavy in a way public-market lending no longer is. This raises a standing question for the rest of the book: when a market's defining feature is the absence of data, how much of the toolkit still applies, and which parts fail first?

3.1.4 Financial Instruments: The Core Contracts

Financial instruments are the contracts traded in markets. Broad categories include debt securities, equity securities, derivatives, and alternative instruments, each with its own economic purpose and pricing logic. Chapter 1's Table 1.4 previewed the markets these instruments trade in; the rest of this section develops each instrument type in depth.

Debt Securities

Suppose a government wants to build a highway that will benefit taxpayers for decades but costs more than this year's tax revenue, or a corporation wants to build a factory that will not turn a profit until it is finished. Both need to raise a large sum now against a promise to repay it, with something extra, over time. Both also need that promise to be tradable, so that whoever lends today is not stuck holding an illiquid IOU for thirty years if their circumstances change. Debt securities solve this problem. A bond is a standardized, tradable loan: the investor lends money to the issuer in exchange for a stream of future payments and the repayment of principal at maturity. Bonds are issued by governments, corporations, and other entities, each with a different credit quality.

Government bonds are often viewed as relatively safe, though they still carry interest-rate and inflation risk. Corporate bonds generally offer higher yields because they are exposed to default risk. Bond pricing rests on the time value of money and discounting: a fixed-coupon bond gains value when market interest rates fall and loses value when they rise, which is why bond prices and yields move inversely.

The bond market also provides a benchmark for the broader economy. Long-term government bond yields are often read as signals of expected growth and inflation, so bond markets are tools for interpreting macroeconomic conditions as well as for financing.

Equity Securities

A bond is a poor fit for a very different kind of firm: one whose value lies almost entirely in an uncertain future, such as a startup that might be worth nothing or billions, with no stable cash flow today from which to make fixed interest payments. A bond's rigid repayment schedule could bankrupt such a firm in a lean year even if its long-run prospects remain excellent. Equity securities let investors share in the firm's fortunes rather than lend against a fixed promise. They are ownership claims: a common stock share gives the holder a claim on future profits and, in many cases, a vote on certain corporate matters. Equity investors are residual claimants, paid only after other obligations such as debt payments are met, so equities are generally riskier than bonds but offer greater upside.

Valuing stocks is harder than valuing bonds because future dividends and firm value are uncertain. Investors use dividend discount models, earnings-based valuation, or relative valuation. In all cases, an equity claim's value depends on expected future cash flows and the discount rate, so a stock promising strong growth may command a high price even if current profits are modest.

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Equity markets also let firms raise permanent capital. Unlike debt, equity has no contractual maturity or scheduled payments. This flexibility suits firms that expect growth, but it means shareholders bear more uncertainty than lenders.

Derivatives

Neither a bond nor a stock lets a wheat farmer lock in today the price of a harvest not yet planted, or lets an airline protect itself against a jet-fuel price spike without buying and storing barrels of oil it has nowhere to put. Both need a contract whose payoff is tied to something else's future price, without either party owning the underlying asset in the meantime. Derivatives are such contracts: their value depends on the performance of another asset or variable. Common examples include forwards, futures, options, and swaps. A forward contract obligates the buyer and seller to exchange an asset at a future date for a predetermined price. A futures contract performs a similar role but is standardized and traded on an exchange. An option gives the holder the right, but not the obligation, to buy or sell an asset at a specified price before or at a certain date.

Derivatives are used for hedging, speculation, and arbitrage. A producer may use futures to lock in a selling price. A portfolio manager may use options to protect against downside risk. A trader may use derivatives to express a view on volatility or market direction. Because derivatives often embed leverage, they magnify gains and losses, which makes them tools for both risk management and risk creation.

By letting parties transfer risk precisely, derivatives expand the risk management strategies available to firms and investors. They can also create hidden exposures if participants misunderstand their payoff structure, one reason regulators and risk managers monitor them closely.

Swaps

Suppose a company has issued floating-rate debt, so its interest payments rise and fall with market rates, but its revenue is stable and it would rather know what it owes each quarter than gamble on rates. Paying off the old loan and negotiating a new fixed-rate one would be slow and expensive, and might be impossible if the lender will not renegotiate. A swap solves this without touching the underlying loan. It is an agreement between two parties to exchange cash flows according to a pre-specified formula, and the most common type is the interest rate swap.

In a plain-vanilla interest rate swap, one party pays a fixed rate on a notional principal amount while the other pays a floating rate, with only the net difference changing hands. The floating rate was historically tied to LIBOR, a benchmark discontinued after regulators found panel banks had manipulated it; transaction-based Alternative Reference Rates such as the Secured Overnight Financing Rate (SOFR) in the U.S. replaced it. The notional principal itself is never exchanged; it only determines the size of the interest payments.

Swaps are used mainly to change the interest rate exposure of an existing position without altering the underlying asset or liability. A company with floating-rate debt that prefers fixed payments can pay fixed and receive floating, converting its liability into a fixed-rate one from a cash-flow perspective. A pension fund matching long-dated fixed liabilities might do the reverse. Currency swaps, credit default swaps (which transfer credit risk, the subject of Chapter 9, rather than interest-rate risk), and total return swaps extend the same idea, exchanging one set of cash flows for another, to other kinds of risk.

Swaps trade almost entirely over-the-counter rather than on exchanges, which historically made their market opaque. Post-2008 reforms, discussed later in this chapter, pushed much of the standardized swap market toward centralized clearing to reduce the counterparty risk concentration that swaps had exposed during the financial crisis.

Worked Example 3.1: Valuing an Existing Interest Rate Swap

A plain-vanilla interest rate swap can be valued as the difference between two bonds. The fixed leg is valued like a coupon bond, discounting each fixed payment plus a final notional repayment. The floating leg's coupon resets to the prevailing market rate, so at any reset date it is worth par (its notional value), whatever interest rates have done since the swap began. Consider a two-year swap with a \$10 million notional and a fixed rate of 5%, agreed some time ago, being valued today, immediately after a reset, when the current one-year and two-year discount rates are 4% and 4.5% respectively, lower than the 5% fixed rate locked in at initiation.

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The fixed leg's annual coupon is $0.05 \times 10{,}000{,}000 = \$500{,}000$, so its present value is

$$PV_{\text{fixed}} = \frac{500{,}000}{1.04} + \frac{500{,}000 + 10{,}000{,}000}{1.045^2} \approx \$10{,}095{,}933.72,$$

while the floating leg, valued at par immediately after its reset, is worth $PV_{\text{floating}} = \$10{,}000{,}000$. The swap's value to the party paying fixed and receiving floating is

$$V_{\text{fixed payer}} = PV_{\text{floating}} - PV_{\text{fixed}} = 10{,}000{,}000 - 10{,}095{,}933.72 \approx -\$95{,}933.72,$$

a loss to the fixed-rate payer and an equal gain to the floating-rate payer: market rates have fallen since the swap was struck, leaving the fixed-rate payer locked into paying 5% when the market now says 4–4.5% is fair. This is how an existing swap's mark-to-market value changes over its life without either counterparty doing anything: movement in market interest rates redistributes value between the two legs. It is the discounting logic Chapter 5 applies to bond pricing, applied to the difference of two cash flow streams.

Other Instruments

Other instruments include money market securities, preferred shares, mortgage-backed securities, and exchange-traded funds. Some are designed for short-term liquidity management, while others combine exposure to many assets in a single tradable instrument. Exchange-traded funds in particular have become widely used because they offer diversification and low-cost access to broad market exposure.

The growth of index funds and exchange-traded funds has changed the structure of investing. Many investors now buy broad baskets that track an index instead of selecting individual securities. This has lowered costs, increased participation, and changed how prices form in some markets, and it shows that modern finance is also about packaging and distributing risk efficiently.

An exchange-traded fund's price is kept close to the value of its holdings through a creation-and-redemption mechanism. Authorized institutional participants can exchange a basket of the underlying securities for new ETF shares, or vice versa, whenever the ETF's market price drifts far enough from its net asset value to make this profitable. This arbitrage, similar in spirit to the no-arbitrage logic in Section 3.4.3, keeps an ETF tracking its index closely in normal conditions, though it can break down under severe stress if the underlying securities become hard to trade.

3.1.5 Interest Rates and Central Banks

Every price in this chapter is quoted against one benchmark: the rate at which money can be borrowed overnight, which a central bank sets. The Federal Reserve announces a target range for the federal funds rate, the rate at which banks lend reserves to one another overnight. It keeps the market rate inside that range mainly by setting the interest it pays on reserves banks hold with it and the rate at which it takes cash from money funds overnight; the European Central Bank's deposit facility rate and the Bank of England's Bank Rate play the same role. No bank will lend overnight for much less than it can earn risk-free at the central bank, and no borrower with good collateral will pay much more than the central bank's lending rate. The policy rate therefore pins the whole overnight complex, including the overnight repo rate worked through later in this chapter and the SOFR benchmark, within a few basis points.

The central bank controls only the shortest rate directly. Longer rates move because the market expects the policy rate to move. Under the *expectations hypothesis*, a two-year yield is the average of the one-year rates expected over the next two years, plus a *term premium* that compensates for the risk of holding the longer bond and that here is set to zero:

$$(1 + y_2)^2 = (1 + r_1)(1 + r_2^e).$$

Suppose the policy rate is 5% and the market expects it to stay there, so $y_2 = 5\%$ and a two-year zero-coupon bond paying \$100 is worth $100/1.05^2 = \$90.7029$. The central bank then surprises with a 50-basis-point cut, and the market comes to expect one more 25-basis-point cut next year, so the expected one-year rates become 4.50% and 4.25%. The two-year yield falls to $\sqrt{1.045 \times 1.0425} - 1 = 4.3749\%$, a move of 62.51 basis points, larger than the cut itself because it also prices next year's expected cut. The zero rises to

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$100/(1.045 \times 1.0425) = \$91.7926$, a gain of 1.2013% (a modified duration of 1.9048 approximates it as 1.1906%, the tool Worked Example 3.3 uses later in this chapter). A stock is a much longer-dated claim. Valued with Chapter 5's Gordon growth model, $P = D_1/(r-g)$, a stock paying $D_1 = \$4$ next year and growing at $g = 4\%$ is worth \$100 at a discount rate of 8%. If the discount rate falls by the same 50 basis points, to 7.5%, the price rises to \$114.29, a gain of 14.29%. Its sensitivity, $1/(r-g) = 25$ years, is more than ten times the two-year zero's, which is why equity markets react so strongly to central bank meetings. There is a caveat: a cut often signals that the central bank expects weaker growth, and if expected dividend growth also falls by 50 basis points, the stock's price does not move at all ($4/(0.075 - 0.035) = \$100$).

Transmission is therefore not mechanical, and the long end can even move the other way. Between September and December 2024 the Federal Reserve cut its target range by a full percentage point, yet the ten-year Treasury yield rose by roughly the same amount, as markets priced higher expected inflation, larger budget deficits, and fewer cuts ahead. With the policy rate near zero, central banks have also worked on the long end directly through *quantitative easing*, buying long-term government and mortgage bonds to compress the term premium. The effect of a policy-rate change depends on what it reveals about the future path of rates, growth, and inflation. That is why Chapter 16 parses central bank statements and Chapter 7 models the whole yield curve rather than a single rate.

3.1.6 Who Does What: The Buy Side, the Sell Side, and Rating Agencies

The participants described so far sort into two camps, a division that also organizes most careers in quantitative finance (Appendix C maps the roles). The **buy side** owns risk on behalf of investors and pays for services: asset managers and mutual funds, pension funds, insurers investing their premiums, sovereign wealth funds, endowments, family offices, and hedge funds. Their quantitative work is portfolio construction, alpha research, and risk management, and their performance is measured against a benchmark. The **sell side** sells services to the buy side and to issuers, and earns fees, spreads, and financing income rather than the return on a portfolio. Its core is the investment bank or broker-dealer, which does four things:

- **Underwriting**: helps issuers sell new stocks and bonds in the primary market, for a fee.
- **Sales and trading**: makes markets for clients in the secondary market, earning the bid-ask spread of Section 3.2.2 and bearing the inventory risk of Section 3.4.1.
- **Research**: publishes analysis of companies and markets, historically paid for through trading commissions.
- **Prime brokerage**: lends cash and securities to hedge funds, holds their collateral, and executes and finances their trades, including through derivatives such as the total return swaps of Section 3.5.4.

The two sides need each other but their incentives differ: a sell-side desk earns when clients trade and borrow, whatever the outcome for the client. Prime brokers therefore compete for a large hedge fund's business partly by offering lower margin, the competition at the center of the Archegos case at the end of this chapter.

A third group sits between issuers and investors. A **credit rating agency** publishes an opinion of an issuer's or a security's relative creditworthiness on a letter scale, from AAA through BBB and down to D (default); securities rated BBB− (Baa3 at Moody's) or higher are *investment grade*, and the rest are high yield, or junk. A rating is an ordinal ranking, not a default probability: it says AAA is safer than A, and only historical default studies, of the kind Chapter 9 uses to map ratings to probabilities, say by how much. Three agencies, S&P Global Ratings, Moody's, and Fitch, issue most ratings, and ratings are built into rules. Many investment mandates may hold only investment-grade bonds, and capital and collateral rules have long referred to ratings, so a downgrade across the investment-grade line can force selling. Since the 1970s the dominant business model has been *issuer pays*: the issuer of the security, not the investor relying on the rating, pays for it. The conflict mattered most for structured products, whose issuers could take a deal to whichever agency's model gave the best rating. The Financial Crisis Inquiry Commission found that from 2000 to 2007 Moody's rated nearly 45,000 mortgage-related securities triple-A, at a time when only six private-sector U.S. companies held that rating, and that 83% of the mortgage securities it rated triple-A in 2006 were ultimately downgraded (Financial Crisis Inquiry Commission 2011). The Dodd-Frank Act responded by creating an SEC Office of Credit Ratings and directing regulators to remove references to ratings from their rules. For model builders the lessons are two: a rating is a feature with a known incentive

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problem attached, and a label such as "AAA" can mean very different things across asset classes even when it is spelled the same.

3.2 Market Microstructure

Zooming in from the market to a single transaction reveals machinery that every later chapter's execution assumptions depend on. This part covers how orders meet, what a large one costs as it walks the book, and where the bid-ask spread comes from.

3.2.1 How Markets Are Organized

Some markets are exchanges with centralized order books; others are over-the-counter markets where participants trade directly with one another. In exchange-traded markets, buyers and sellers submit orders that are matched according to price and time priority. In over-the-counter markets, contracts are negotiated bilaterally and may be tailored to specific needs.

The distinction affects liquidity, transparency, counterparty risk, and transaction costs. Exchange-traded products are often easier to price and monitor, while over-the-counter trades offer flexibility but require more careful risk management. An exchange-listed stock can usually be traded quickly at a visible price, whereas a bespoke derivative contract may require negotiation, credit assessment, and collateral arrangements.

The study of these rules is called market microstructure. Even small changes in execution rules, fees, or order types can affect the quality of price discovery. A limit order may obtain a better price but may fail to execute. A market order executes immediately but at the cost of adverse price impact. These trade-offs are central to trading and to research in finance.

A simple measure of trading cost is the bid-ask spread:

$$\text{Spread} = P_{\text{ask}} - P_{\text{bid}}. \tag{3.1}$$

The spread is the cost of immediacy: a narrow spread means cheap trading and strong liquidity, a wide one the reverse. For quantitative researchers it is often as important as the asset's return, because it determines whether a strategy is profitable in practice.

3.2.2 Worked Example 3.2: Walking the Limit Order Book

The bid-ask spread describes the price of trading a single share. A limit order book lists several price levels on both sides, and a large order can "walk" through them, receiving a progressively worse average price. Table 3.1 shows a simplified order book for a stock.

Asks (sell orders)			Bids (buy orders)		
Level	Price	Size (shares)	Level	Price	Size (shares)
1	\$50.10	100	1	\$50.05	150
2	\$50.15	200	2	\$50.00	250
3	\$50.20	300	3	\$49.95	200

Table 3.1: A simplified limit order book

The best bid is \$50.05 and the best ask is \$50.10, so the quoted bid-ask spread is \$0.05, consistent with Equation (3.1). Figure 3.1 draws the same book as a depth chart, the standard way a trading desk visualizes it. Now suppose a trader submits a market buy order for 250 shares, more than the 100 shares available at the best ask. The order first consumes all 100 shares at \$50.10, then the remaining 150 shares at the next level, \$50.15:

$$\text{Average execution price} = \frac{100(50.10) + 150(50.15)}{250} = \frac{5{,}010 + 7{,}522.50}{250} = \$50.13.$$

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This average price is \$0.03 above the best ask of \$50.10, roughly 6 basis points of price impact, a consequence of order size relative to the liquidity at each price level, with no new information entering the market. A trader with a large position faces a trade-off: submitting the whole order at once guarantees immediate execution at a worse average price, while breaking it into smaller pieces over time (the execution algorithms of Chapter 13) reduces price impact but exposes the trader to price movements while the order is worked.

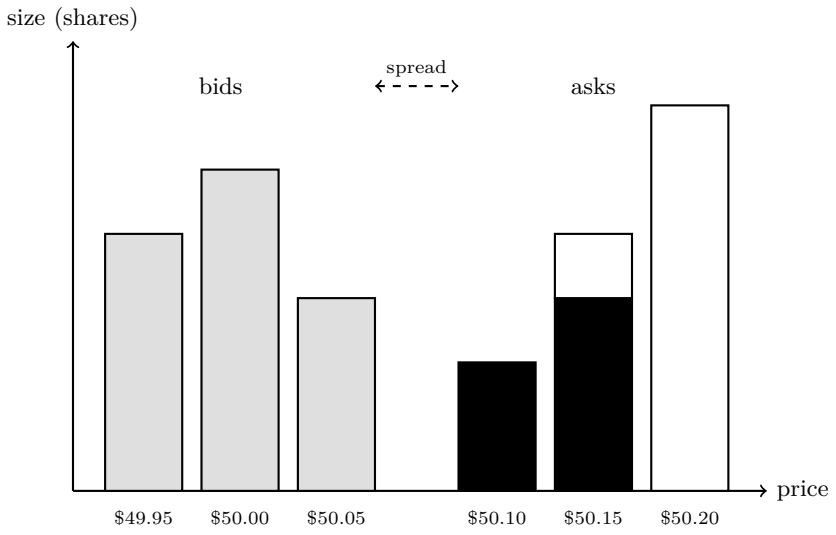


Figure 3.1: The limit order book of Table 3.1 as a depth chart. Bid sizes (light gray) sit left of the spread, ask sizes (white outline) right; the solid black portion of the two lowest ask levels marks what the 250-share market order consumes.

3.2.3 Decomposing the Bid-Ask Spread

The bid-ask spread is not pure profit for a market maker; it compensates for at least three distinct costs, and the decomposition explains why spreads widen in some conditions and narrow in others.

- **Order-processing cost**: the operational cost of handling a trade, which is largely fixed regardless of market conditions.
- **Inventory-holding cost**: compensation for the risk a market maker bears by holding a nonzero position while waiting for an offsetting order to arrive, discussed further in Section 3.4.1.
- **Adverse-selection cost**: compensation for the risk of trading with a counterparty who has better information, since an informed trader will systematically pick off stale quotes.

A trade price bounces between the bid and the ask as buy and sell orders arrive, even when the asset's underlying value has not changed. A trade at the ask followed by a trade at the bid looks, in the raw price series, like the price falling and then recovering, so a wider spread mechanically produces larger back-and-forth price changes. Roll's implied spread estimator (Roll 1984) turns this bounce into a measurement tool, recovering the effective spread from observed price changes alone, without seeing the order book,

$$\text{Spread} \approx 2\sqrt{-\text{Cov}(\Delta P_t, \Delta P_{t-1})}, \tag{3.2}$$

because this bid-ask bounce induces negative autocorrelation in consecutive price changes, which the covariance term captures. This is a useful reminder for the time series methods of Chapter 7: not every statistical pattern in price data reflects economically meaningful predictability; some of it is a mechanical consequence of market microstructure. The book's companion technical note on Roll's estimator derives the formula. It shows that a constant half-spread applied according to the direction of each arriving trade makes observed

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price changes behave like a first-order moving-average process, with an autocovariance of $-s^2/4$ at lag one and zero at every longer lag, and that bounce alone can never drive the first-lag autocorrelation below $-1/2$ however wide the spread.

3.2.4 The Economics of Price Discovery

Financial prices are the output of a process in which participants with different information and objectives update their beliefs. A stock price may rise because investors revise their expectations of future cash flows, because risk premiums change, or because a large buyer enters the market. The same price change can have several interpretations, which is why finance is both quantitative and interpretive.

Because information, incentives, and trading frictions all affect prices, they do not always move smoothly. Sudden moves can reflect news, uncertainty, algorithmic trading, or liquidity shocks. In the short run, prices may deviate from fundamental values because of temporary imbalance or limited liquidity; in the long run, competition and arbitrage tend to push them toward values consistent with available information and fundamentals.

Economists and practitioners therefore distinguish fundamentals from short-run market dynamics. A fundamental analyst asks what an asset ought to be worth given expected cash flows and risk. A market microstructure analyst asks how trading rules and order flow shape the observed price path. Both perspectives are needed.

Price discovery also matters for AI applications. A model that finds patterns in observed prices without understanding the market mechanism behind them may mistake noise for signal, and a model that predicts price movements may still fail if it ignores transaction costs, market impact, and changing liquidity.

3.2.5 Why Market Structure Matters for Quantitative Finance

A portfolio manager does not merely choose assets by expected return and risk. The manager must also consider how easily those assets can be bought or sold, how much price impact trading will create, and whether the strategy survives transaction costs.

The same logic applies to credit risk models, fraud detection systems, and forecasting algorithms. A model that performs well on historical data may fail in practice if it ignores the institutional setting in which decisions are implemented. Finance is a problem of data, incentives, constraints, and execution, and the most successful applications of AI combine rigorous modeling with an understanding of the market mechanism rather than treating markets as a black box.

Suppose a machine learning model, trained only on historical price and volume data, predicts that a small-cap stock will outperform over the next week.

Before acting on this signal, a sophisticated practitioner asks institutional questions the model cannot answer. How wide is the bid-ask spread, and does the expected outperformance exceed the round-trip cost of trading, using the spread decomposition above? What share of the stock's average daily volume would the position represent, and would building it through the order-book mechanics of Section 3.2.2 move the price enough to erode the gain? Is the stock easy to borrow if the strategy requires shorting, and are there restrictions on shorting small-cap names under stress?

None of these questions is visible in the price data the model was trained on, yet each can determine whether a statistically significant signal is also economically exploitable. Market structure, not just statistical significance, decides whether a trading idea survives contact with a real market.

3.3 Market Infrastructure and Regulation

A trade is not finished when it is agreed. This part follows it through clearing and settlement, examines the businesses and rules that surround it, and connects that plumbing to where machine learning is deployed.

3.3.1 Money Markets, Capital Markets, and the Lifecycle of a Trade

Financial markets are often divided into money markets and capital markets. Money markets are used for short-term borrowing and lending, typically for maturities of less than one year. They include Treasury bills, commercial paper, certificates of deposit, and repurchase agreements, and are crucial for liquidity

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management, since banks, corporations, and governments often need to finance short-term obligations or invest excess cash briefly. Chapter 1's Table 1.4 previewed money markets alongside equity, bond, derivatives, and foreign exchange markets; this section and Chapter 5 develop each of these further.

Capital markets, by contrast, are used for long-term financing. They include equity and long-term debt markets and fund business investment and public infrastructure. A firm expanding production may issue long-term bonds or equity; a government may issue sovereign debt to finance roads, schools, or defense. In both cases, the capital market separates the financing decision from the issuer's immediate cash position.

Another distinction is between primary and secondary markets. In a primary market, new securities are issued for the first time. In a secondary market, existing securities are traded among investors. The primary market raises funds for the issuer, while the secondary market sets the price of securities after issuance. A liquid secondary market reduces uncertainty for investors and so makes it easier for firms to issue securities in the first place.

The lifecycle of a trade shows why market structure matters in practice. An investor decides to buy or sell an asset, and the order is routed through a broker to an exchange or an over-the-counter dealer. In an electronic, highly liquid market the order may execute almost immediately; in a fragmented or less liquid one it may be partially filled or require price concessions. After the trade, settlement transfers ownership and exchanges cash. In many markets settlement takes a few days, during which counterparty and operational risk remain.

The process reflects the interaction of incentives, technology, and regulation. High-frequency trading firms may compete on execution speed, while institutional investors prioritize minimizing market impact. A retail investor may value simplicity and low cost, whereas a hedge fund may bear more risk for a strategic advantage. All of these objectives interact in the same marketplace.

The repurchase agreement, or repo, deserves particular attention because of its central role in the plumbing of the financial system. In a repo, one party sells a security, typically a government bond, to another while agreeing to repurchase it at a slightly higher price on a specified future date, often the next day. Economically, this is a collateralized short-term loan: the cash borrower posts the security as collateral, and the difference between the sale and repurchase price is the effective interest rate, known as the repo rate. Banks, hedge funds, and other institutions rely on the repo market to fund large securities positions daily. A disruption in repo markets, as occurred at several points during the 2008 crisis, can therefore force institutions to sell assets they would otherwise have held, transmitting stress across the financial system as described in the systemic risk discussion below.

Suppose a dealer needs to borrow \$10,000,000 overnight and posts Treasury securities as collateral in exchange for cash at an annualized overnight repo rate of 5%. Using the money market convention of a 360-day year, the interest owed for one day is

$$\$10{,}000{,}000 \times 0.05 \times \frac{1}{360} = \$1{,}388.89,$$

so the dealer repurchases the collateral the next day for \$10,001,388.89. This small daily interest, applied to an enormous notional and rolled over every business day, is why even a small rise in overnight repo rates can signal a meaningful shortage of cash for short-term collateralized lending, as in September 2019, when repo rates briefly spiked well above the Federal Reserve's target range.

Settlement conventions also matter in practice. Most equity markets settle trades on a T+1 or T+2 basis (one or two business days after the trade date), so ownership and cash do not change hands when a trade is agreed. This gap creates counterparty risk, the possibility that the other side fails to deliver cash or securities as promised, which the clearinghouses discussed below exist to mitigate.

3.3.2 The Business of Exchanges and Market Data

It is easy to think of an exchange as neutral infrastructure, a venue where buyers and sellers meet, without asking how it makes money. Modern exchanges are for-profit businesses with several revenue streams, and these explain features of market structure that would otherwise seem arbitrary.

Listing fees are charged to companies for the right to have their shares traded on an exchange, giving exchanges an incentive to attract high-quality, high-volume issuers. Transaction fees are charged per trade, often under a "maker-taker" model in which an exchange pays a small rebate to participants who post resting

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limit orders (adding, or "making," liquidity) and charges a slightly larger fee to participants whose orders execute immediately against those resting orders (removing, or "taking," liquidity). This fee structure is a deliberate subsidy for market making, and it materially affects the profitability of the market makers discussed in Section 3.4.1.

Market data fees are a third, increasingly significant revenue source: exchanges sell real-time price and order-book data to trading firms, and the most detailed, lowest-latency feeds command large premiums over the free or delayed data available to retail investors. Finally, co-location fees, charged for placing a firm's servers inside or next to the exchange's data center, monetize the speed advantage discussed later in this chapter. A co-located firm receives order-flow information and executes its orders with lower latency than a remote firm, purely because data travels a shorter physical distance.

These revenue streams matter to a quantitative reader for two reasons. First, they explain why exchanges have commercial incentives to attract high-frequency trading in particular, which generates outsized transaction and data-fee revenue relative to its economic footprint; this feeds a broader regulatory debate about whether market structure balances the interests of high-frequency participants against those of long-term investors. Second, the market data used throughout this book, assumed free in every worked example so far, is a commercial product whose cost, coverage, and latency depend on how much a firm will pay, a constraint that shapes which strategies a research budget can support.

3.3.3 Clearing, Settlement, and Systemic Risk

Once a trade is executed, the market must ensure that it is honored. Clearing determines the obligations between counterparties; settlement is the final exchange of cash and securities. In exchange-traded markets, a clearinghouse often acts as the central counterparty, guaranteeing the trade if one side defaults. This reduces bilateral credit risk, but it also creates a new concentration of risk in the clearinghouse itself.

Clearing and settlement matter most under stress. If many participants must liquidate positions at once, the system may face liquidity shortages and margin calls, and a small shock can propagate through the network of institutions that depend on one another for financing and collateral. This is the core idea of systemic risk: the failure of one institution or market segment can quickly undermine the stability of the broader financial system.

The 2008 global financial crisis showed how quickly seemingly isolated problems become systemic. U.S. mortgage defaults spread through securitization structures, derivatives markets, and funding markets, eventually affecting banks, insurers, and governments worldwide. The lesson was not only that risk had been mispriced, but that the financial system had become highly interconnected and insufficiently transparent. The crisis also highlighted the importance of rules on leverage, capital, and liquidity.

A central clearinghouse manages its concentrated risk mainly through margin requirements. Each clearing member posts collateral, called initial margin, sized to cover the clearinghouse's potential loss if that member defaults under plausible adverse price moves, and posts variation margin daily (or, in volatile periods, intraday) to cover losses on open positions. This protects the clearinghouse, but a sharp price move can trigger margin calls across many participants at once, forcing coordinated selling when the market is least able to absorb it, a dynamic visible in several episodes of the 2008 crisis. Major clearinghouses have become large and interconnected enough with their member banks that regulators now treat them as systemically important institutions, subject to their own capital and risk-management standards.

Settlement is also where new market plumbing has appeared, in the form of stablecoins and tokenized securities. Both are best understood alongside the systems that already move money, so the next subsection takes them up together.

3.3.4 Payment Infrastructure and Digital Assets

The same three questions asked of securities clearing (who owes whom, when an obligation becomes final, and who bears the risk in between) arise every time money moves. This subsection looks first at the payment rails that move money today, then at the digital assets, stablecoins and tokenized securities, that propose to move money and securities together on a new kind of ledger, and closes briefly with central bank digital currencies. The subject is plumbing and its risks. Crypto-asset trading strategies are outside this book's scope, as the preface states, and nothing here is an argument for or against holding any digital asset.

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How Payments Settle

Payment systems differ along two dimensions that matter for risk: whether each payment settles individually (gross) or in batches netted against one another, and how long a payment takes to become final, meaning legally irrevocable. Table 3.2 summarizes the four main designs.

A card payment is authorized in seconds: the merchant's bank (the acquirer) sends a message through the card network to the cardholder's bank (the issuer), which checks the available balance or credit line and a fraud score before approving. The money moves later. Transactions are cleared in batches and settled between issuer and acquirer a day or two afterward, and the cardholder keeps the right to dispute a charge, a chargeback, for weeks. Card payments are therefore fast to approve but slow to become final, and that delayed finality makes card-fraud losses at least partly recoverable, the setting for much of Chapter 15. The automated clearing house (ACH) network in the United States, which carries payroll, bill payments, and direct debits, is batch-based too: banks submit files of payments, the operator nets them, and settlement occurs at fixed windows during the business day. Netting economizes on liquidity, since each bank funds only its net position, but a payment remains a promise until its batch settles, and some debits can be returned days later.

Large-value payments between banks instead settle through real-time gross settlement (RTGS) systems operated by central banks: the Fedwire Funds Service in the United States, CHAPS in the United Kingdom, and T2 in the euro area. Each payment settles individually and immediately in central bank money and is final the moment it settles. The price of that safety is liquidity: with no netting, a bank must hold or borrow enough reserves to fund every payment as it goes, which is why central banks extend intraday credit to RTGS participants.

Instant payment rails bring RTGS-style finality to everyday payments. The Clearing House's RTP network (launched in 2017) and the Federal Reserve's FedNow Service (launched in July 2023) settle individual payments within seconds, around the clock and every day of the year, as do the United Kingdom's Faster Payments and the euro area's SEPA Instant Credit Transfer scheme. Finality cuts both ways: a firm paying a supplier wants it, and so does a fraudster. In an authorized push payment (APP) scam, the victim is manipulated into sending the money personally, against a fake invoice or at the request of someone impersonating the bank. Once the payment settles on an instant rail there is no chargeback to reverse it, and the funds are typically moved onward through mule accounts within minutes. In the United Kingdom, where APP losses on Faster Payments became a policy issue, sending and receiving payment firms have since 7 October 2024 been required to reimburse most victims, up to £85,000 per claim. Chapter 15 treats APP fraud as a detection problem; the structural point here is that the faster and more final a rail, the more of the fraud screening must happen in the seconds between the payment instruction and settlement, because nothing can be clawed back afterward.

Rail	Settlement	When final	Typical use
Card networks	Batched, net	Days; disputable later	Retail purchases
ACH	Batched, net	Same day to two days	Payroll, bills, direct debits
RTGS (Fedwire, CHAPS, T2)	Individual, gross	Immediately	Large interbank payments
Instant (RTP, FedNow)	Individual, gross	Seconds, around the clock	Person-to-person, business

Table 3.2: Four payment rails compared by how they settle and when a payment becomes final

Stablecoins: Reserves, Redemption, and Run Risk

A stablecoin is a token on a blockchain designed to hold a fixed value, almost always $1. It does so by being redeemable on demand at par from an issuer that holds reserves of cash and short-term Treasuries against the tokens outstanding. (Algorithmic stablecoins, which tried to hold the peg without full reserves, are a different design, largely discredited by the collapse of TerraUSD in May 2022; this subsection concerns the fiat-backed reserve model that now dominates and that regulators have adopted.) A fiat-backed stablecoin is economically a money-market fund share that gives up yield in exchange for functioning as money. The issuer keeps the interest earned on the reserves, which is its business model, and the holder gets a token that moves

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around the clock on a blockchain and can serve as the cash leg of the tokenized-securities trades described below. The quality of its reserves is the credit question Chapter 9 would ask of any issuer.

The promise of redemption at par is also the source of fragility. A holder who doubts the reserves can redeem at \$1 today; if enough holders do, the issuer must sell reserve assets to pay them, possibly at a loss, and the loss falls on those who wait. This is the structure of a bank run (Diamond and Dybvig 1983), and stablecoins have "broken the buck" in this way. In March 2023 the issuer of USDC disclosed that \$3.3 billion of its roughly \$40 billion of reserves was held at Silicon Valley Bank, which had just failed. The token traded below 90 cents before recovering within days, once U.S. authorities guaranteed that bank's deposits.

Regulation has converged on one design, as of October 2026. In the United States, the GENIUS Act, signed into law on July 18, 2025, requires a permitted payment stablecoin issuer to hold identifiable reserves on at least a one-to-one basis, limited to cash, insured deposits, Treasury bills with a remaining maturity of 93 days or less, Treasury-backed repurchase agreements, government money-market funds, and central bank reserves; to publish the composition of its reserves monthly, examined by a registered public accounting firm; to disclose a redemption policy providing for timely redemption; and not to pay holders any interest or yield (U.S. Congress 2025). The Act takes effect on the earlier of January 18, 2027 (eighteen months after enactment) and 120 days after the federal regulators issue final implementing rules, which as of October 2026 had been proposed but not finalized. In the European Union, the Markets in Crypto-Assets Regulation (MiCA) applied its stablecoin provisions from 30 June 2024 and the rest of the regulation from 30 December 2024. It gives holders of an e-money token, the MiCA category for a stablecoin tied to a single official currency, the right to redeem at any time and at par, bars issuers from paying interest, and requires that at least 30% of the funds received be held as deposits at credit institutions, with the remainder in highly liquid, low-risk instruments (EU 2023). Full reserves, short maturities, redemption at par, and no interest: the next worked example shows why the second of these matters.

Worked Example 3.3: A Stablecoin Reserve Under a Redemption Run

An issuer has $S = \$10{,}000$ million of tokens outstanding, each redeemable at \$1, backed one for one by $C = \$1{,}500$ million of cash (15%) and $B = \$8{,}500$ million of zero-coupon Treasuries bought at a yield of $y = 4\%$. At that yield the Treasuries earn the issuer $0.04 \times 8{,}500 = \$340$ million a year, none of which is paid to holders. Two things then happen: yields rise by $\Delta y = 100$ basis points, and 30% of holders redeem, $R = \$3{,}000$ million.

Measuring the damage requires one tool from fixed income, developed fully in Chapter 5 and needed here only for the simplest bond. A zero-coupon bond paying \$1 in T years is worth $P = (1+y)^{-T}$, so $dP/dy = -\frac{T}{1+y}P$, and for a small yield change

$$\frac{\Delta P}{P} \approx -D_{\text{mod}}\,\Delta y, \qquad D_{\text{mod}} = \frac{T}{1+y}, \tag{3.3}$$

where D_{mod} is the bond's modified duration: the percentage price loss per unit rise in yield. Compare two reserve portfolios. A compliant one holds three-month bills, $T = 0.25$, inside the GENIUS Act's 93-day limit: $D_{\text{mod}} = 0.25/1.04 = 0.2404$, so the bills lose about 0.2404% (exact repricing gives 0.2390%), a mark-to-market loss of $L = 8{,}500 \times 0.002404 = \$20.43$ million. An extended one holds two-year zeros, $T = 2$: $D_{\text{mod}} = 2/1.04 = 1.9231$, a loss of about 1.9231% (exact 1.8957%), or $L = \$163.46$ million, eight times as much.

The redemption is paid first from cash and then by selling Treasuries. The issuer needs $3{,}000 - 1{,}500 = \$1{,}500$ million from sales, and at marked-down prices that means selling Treasuries that cost $1{,}500/(1 - 0.002404) = \$1{,}503.61$ million in the compliant case (a realized loss of \$3.61 million) and $1{,}500/(1 - 0.019231) = \$1{,}529.41$ million in the extended case (a realized loss of \$29.41 million). The redeeming holders are paid exactly \$1 each, so the whole mark-to-market loss L, realized or not, is left to the $S - R$ tokens still outstanding. Reserves per remaining token are

$$c = \frac{S - L - R}{S - R} = 1 - \frac{L}{S - R}, \tag{3.4}$$

which gives $1 - 20.43/7{,}000 = \$0.9971$ for the compliant issuer and $1 - 163.46/7{,}000 = \$0.9766$ for the extended one. Table 3.3 repeats the calculation as the redeemed share grows.

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Share redeemed	Three-month bills ($T = 0.25$)	Two-year zeros ($T = 2$)
0%	0.9980	0.9837
15%	0.9976	0.9808
30%	0.9971	0.9766
60%	0.9949	0.9591

Table 3.3: Reserves per remaining token, in dollars, after a 100-basis-point rise in yields, by share of tokens redeemed at par

Equation (3.4) is the arithmetic of a run. Every holder who redeems early receives a full dollar, and the same loss is spread over fewer remaining tokens, so the value left behind falls as redemptions grow: the extended issuer's remaining holders have 98.37 cents of reserves per token before anyone redeems and 95.91 cents after 60% have left. A holder who expects others to redeem should therefore redeem first, and the expectation can fulfill itself. The 93-day limit does not remove this first-mover advantage; it shrinks it by the ratio of the durations, $1.9231/0.2404 = 8$, so that the compliant issuer's worst case in the table is about half a cent below par, small enough for a modest capital buffer to absorb. A larger cash buffer, such as MiCA's 30% minimum, lets the issuer meet a redemption without selling anything, which avoids the realized loss, but the economic loss on the Treasuries still held is unchanged by the decision not to sell. The same arithmetic, applied to a bank funding long-duration securities with runnable deposits, is what brought down Silicon Valley Bank in 2023. The first-order approximation slightly overstates the losses (compare 1.9231% with the exact 1.8957%), a convexity effect explained in Chapter 5.

Tokenized Securities and Money-Market Funds

Asset tokenization means issuing a conventional security (a money-market fund share, a Treasury bill, a bond) as a digital token whose ownership record is a blockchain rather than an entry at a custodian or transfer agent. The motivation is settlement economics, not speculation. A tokenized security can settle in minutes rather than days, with delivery and payment executed atomically in a single transaction so that neither leg can fail on its own, collapsing both the intermediation cost and the counterparty-risk window explained earlier. Atomic settlement requires both legs to live on the same rails, which is where a stablecoin or tokenized bank deposit supplies the cash leg. Tokenized money-market funds are the most developed case: BlackRock's USD Institutional Digital Liquidity Fund (BUIDL), launched on the Ethereum blockchain in March 2024, invests in cash, Treasury bills, and repurchase agreements and records each investor's shares as tokens that can be transferred around the clock and, in principle, posted as collateral while still earning the fund's yield.

Tokenization changes the settlement cycle, the hours during which an asset can move, and the possibility of attaching rules (who may hold the token, when it pays out) directly to the asset as code. It does not change the asset's economics: a tokenized Treasury bill has the same duration and credit risk as the bill, and the run arithmetic of Equation (3.4) applies to any token promising par redemption against assets that can fall in value. It does not by itself settle legal questions, since a holder's claim is only as good as the law's recognition of the token record and the issuer's redemption terms. And atomic settlement gives up something that batch settlement provided: because each trade must be fully funded at the moment it settles, the liquidity savings of netting disappear, the same trade-off between safety and liquidity that separates RTGS from ACH.

For a quantitative reader the consequence is concrete. A settlement cycle collapsing from days to minutes changes collateral requirements, intraday liquidity management, and the practical meaning of counterparty exposure, all of which feed the risk models of Chapters 8 and 9. The venues where tokenized instruments trade also have formula-driven rather than dealer-driven market making, taken up in Chapter 13.

Central Bank Digital Currencies

A central bank digital currency (CBDC) is a digital form of money that is a direct liability of the central bank, held either by the public (a retail CBDC) or only by banks for settling with one another (a wholesale CBDC). Because it is central bank money, a CBDC has no reserve portfolio and none of the run arithmetic above.

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The concern runs the other way: in a crisis households could move deposits out of banks and into CBDC at the click of a button, which is why proposals such as the digital euro limit individual holdings. Policy has diverged, as of October 2026. China has piloted its retail e-CNY widely; the Eurosystem has been preparing a possible digital euro since November 2023, with any issuance conditional on EU legislation; and in the United States an executive order of January 23, 2025, prohibits federal agencies from establishing, issuing, or promoting a CBDC, leaving regulated stablecoins and tokenized deposits as the U.S. route to digital dollars. Central banks themselves increasingly emphasize wholesale uses, tokenized central bank reserves settling against tokenized deposits and government bonds on a shared ledger (Bank for International Settlements 2025).

For this book, the data matter as much as the designs. Instant payment rails produce the real-time transaction streams that Chapter 15's models must score before a payment becomes final, and public blockchains publish every token transfer, the on-chain data that Chapter 17 treats as an alternative data source.

3.3.5 Regulation, Disclosure, and Market Integrity

Markets need rules because they rest on trust, information, and incentives. Without regulation, participants may exploit others through fraud, insider trading, or manipulation. Disclosure requirements help investors evaluate firms and make informed decisions. Capital and liquidity rules reduce the chance that institutions will become insolvent or unable to meet obligations during stress. Market surveillance systems detect suspicious behavior and support confidence in the fairness of the trading process.

The balance between free markets and regulation is a persistent theme. Too little regulation may encourage excessive risk-taking and misconduct; too much may reduce innovation and impose unnecessary compliance costs. Effective frameworks protect integrity and stability while preserving the ability of markets to allocate capital efficiently.

For AI and data science, regulation also shapes the data environment. Financial institutions must report certain information, comply with auditing standards, and keep transparent records, which creates rich datasets for forecasting, fraud detection, and risk analysis. Regulatory constraints also limit what can be done with the data, especially where privacy, fairness, or market abuse is involved.

A few regulatory frameworks recur throughout the book and are worth knowing by name.

- **The Securities and Exchange Commission (SEC)** oversees U.S. securities markets and disclosure.
- **Regulation NMS** (National Market System) governs how orders are routed and executed across competing U.S. exchanges, directly shaping the market microstructure discussed throughout this chapter.
- **The Dodd-Frank Act**, passed in 2010 in response to the 2008 financial crisis, introduced the central clearing requirements for standardized swaps mentioned above, along with the Volcker Rule restricting proprietary trading by banks, and stress-testing requirements for large financial institutions that are directly relevant to the risk models in Chapters 8 and 9.
- **MiFID II** (the Markets in Financial Instruments Directive), in Europe, imposes extensive transparency and best-execution requirements on trading venues and investment firms.
- **Basel III** is an international framework setting minimum capital and liquidity requirements for banks, intended to make the banking system more resilient to the kind of interconnected shocks described in the systemic risk discussion above.

None of these frameworks is a topic this book covers in depth, but recognizing their names and general purpose is useful context for the compliance-related material in Chapter 15.

3.3.6 The Link Between Market Structure and AI

A modern AI system in finance cannot be designed as if markets were simple, frictionless, and fully observable. Real markets have heterogeneous participants, delayed information, hidden inventory, and strategic behavior. An algorithm that trades on price patterns may perform well in a calm market but fail under stress, when liquidity evaporates and spreads widen. A credit model built on historical repayment data may become less reliable if underwriting standards change or the economy enters a new regime.

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Financial machine learning is therefore an engineering problem embedded in an economic environment. The model is one component of a system that includes data pipelines, risk controls, execution logic, and compliance procedures, and the same prediction can have very different consequences depending on how it is used. A recommendation harmless in research may be dangerous in live trading if it creates concentration risk or triggers herding.

The most valuable AI applications in finance respect the structure of the market: they use data carefully, incorporate financial theory, and are judged by their economic and operational consequences as well as their predictive accuracy.

The connection also runs the other way: AI systems have become part of the market structure they operate within.

The high-frequency market-making firms discussed later in this chapter increasingly use machine learning to set quotes and manage inventory risk in real time. Large asset managers use natural language processing (Chapter 16) to parse earnings calls and regulatory filings faster than human analysts. Exchanges use anomaly-detection systems, related to the fraud-detection methods of Chapter 15, to flag potentially manipulative trading for regulatory review. The "other participants" whose behavior shapes liquidity, spreads, and price discovery are thus increasingly AI systems, whose collective behavior any new AI system entering the market must anticipate.

This has a subtle consequence for model validation. A backtest evaluates a strategy against data generated by the market as it existed in the past, including the mix of human and automated participants active then. If much of that liquidity and price-setting came from algorithms since retired, upgraded, or replaced, the market a new strategy faces may behave differently from its backtest, not because fundamentals changed but because the population of participants generating prices has changed. This is a subtler version of the non-stationarity concern raised in Chapter 1 and developed in Chapter 6, specific to markets with heavy algorithmic participation.

3.4 Liquidity, Leverage, and Arbitrage

Prices hold together because someone is willing to trade against a discrepancy. This part covers who provides that liquidity and on what terms, what borrowing to trade costs when it goes wrong, and the no-arbitrage logic the rest of the book prices with.

3.4.1 Liquidity, Leverage, and the Economics of Market Making

Liquidity is one of the most important but least visible features of financial markets. A liquid market allows participants to trade quickly without moving the price much. Liquidity has several dimensions:

- **Depth** is the volume that can be traded at a given price.
- **Immediacy** is the speed with which an order can be executed.
- **Resiliency** is the speed with which the market returns to a stable state after a shock.
- **Tightness** is the narrowness of the bid-ask spread.

A liquid market is easier to use for hedging, investment, and risk management, because traders can enter and exit positions more cheaply.

Market makers do more than passively post a bid and an ask: they absorb temporary imbalances between buy and sell orders, buying when others want to sell and selling when others want to buy. This is where inventory risk bites. If the market maker buys an asset and its price falls, the position loses value; if it sells short and the price rises, losses can also accumulate quickly. The willingness of market makers to bear this risk is central to market functioning.

Market making is also shaped by information asymmetry. When some traders have better information, market makers face adverse selection. An informed trader who expects an asset to rise is more likely to buy, and one who expects it to fall is more likely to sell. The market maker, who cannot tell whether a trade conveys good or bad news, may respond by widening spreads and reducing participation, one reason trading costs often rise in uncertain periods.

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Leverage is another essential concept here. A trader who borrows to increase position size magnifies both gains and losses. Margin requirements limit how large a leveraged position can become relative to the trader's equity. When prices move against the position, a margin call may force liquidation, and that forced selling can amplify price moves and add stress to the market. Leverage, useful for efficient investment and speculation, is therefore also a source of instability when risk management is weak.

3.4.2 Worked Example 3.4: Short Selling and Margin Calls

Leverage and margin are easiest to understand through a short sale. Suppose a trader believes a stock currently priced at \$50 is overvalued and borrows 100 shares to sell short, receiving proceeds of $100 \times \$50 = \$5{,}000$. Under a typical initial margin requirement of 150% of the value of the position (the U.S. Regulation T standard), the trader must maintain total account equity of at least $1.5 \times \$5{,}000 = \$7{,}500$: the \$5,000 in short-sale proceeds plus an additional \$2,500 of the trader's own cash deposited as margin.

As the stock price P moves, the trader's liability, the cost of buying back the 100 borrowed shares, is $100P$, while total account equity remains the fixed \$7,500 credit balance minus that liability:

$$\text{Equity}(P) = 7{,}500 - 100P.$$

For short positions in stocks priced at \$5 or more, FINRA Rule 4210 requires a maintenance margin of at least 30% of the current value of the short position (many brokers require more), meaning $\text{Equity}(P)/100P \geq 0.30$. Setting this ratio equal to 0.30 and solving for P gives the price at which a margin call is triggered:

$$7{,}500 - 100P = 0.30(100P) \quad \Longrightarrow \quad 7{,}500 = 130P \quad \Longrightarrow \quad P \approx \$57.69.$$

At this price (7,500/130 at full precision), equity has fallen to \$1,730.77, exactly 30% of the trader's liability, confirming the calculation. A price increase of only 15.38%, from \$50 to \$57.69, triggers a margin call on a position that started with 150% initial margin, which shows how leverage compounds a short seller's losses. The loss on the position is \$769.23, but it consumes 30.77% of the trader's own \$2,500 of capital, twice the adverse price move in percentage terms. If the trader cannot deposit more funds once the call is issued, the broker buys back some or all of the borrowed shares on the trader's behalf at whatever price is available. This is the same price-insensitive trading into a moving market that the Flash Crash vignette describes and, at the scale of a whole leveraged fund, the LTCM episode discussed later in this chapter.

Variation margin, the clearinghouse mechanism of Section 3.3.3, differs from broker margin in one important respect: rather than being called only once a position breaches a maintenance threshold, it is settled at the end of every trading day, marking every open position to its current market price.

Suppose a trader buys one stock-index futures contract with a notional value of \$50,000 and posts an initial margin of \$5,000 (10% of notional). If the index falls 2% that day, the trader's mark-to-market loss is $0.02 \times \$50{,}000 = \$1{,}000$, reducing the margin balance to $\$5{,}000 - \$1{,}000 = \$4{,}000$; the clearinghouse issues a variation margin call for \$1,000 to restore the balance to \$5,000 before trading resumes the next day. Because settlement is daily, losses on a futures position never accumulate silently as they can in the short-sale example. But a clearing member with many leveraged futures positions faces a new, potentially large cash obligation every time the market moves sharply against it, the liquidity drain described in Section 3.3.3.

Table 3.4 compares the two margin mechanics, which are easy to conflate despite their different timing and triggers.

	Short-sale margin (Section 3.4.2)	Futures variation margin
Counterparty	Broker	Central clearinghouse
Settlement frequency	Only when a threshold is breached	Every trading day
Trigger	Equity falls below maintenance %	Any price move, however small
Typical use case	Individual equity short position	Standardized futures and swaps

Table 3.4: Two margin mechanisms compared: bilateral broker margin on a short sale versus centrally cleared futures margin

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3.4.3 A Simple No-Arbitrage Example

A simple arbitrage argument shows why prices in finance are constrained by structure. Suppose an investor can buy a one-period risk-free bond that costs 100 and pays 105 one year later. The risk-free interest rate implied by this bond is

$$r = \frac{105 - 100}{100} = 0.05.$$

Now suppose there is also a claim that pays one unit of money with certainty in one year. If the claim were priced at 0.95 today, then an investor could buy the claim for 0.95 and receive 1 next year. This is a return of about 5.26%, higher than the risk-free rate available from the bond, so rational investors would buy the claim and sell the bond, pushing prices back into line. The no-arbitrage principle says that two assets with the same payoff must have the same price; otherwise investors can make a riskless profit. This idea underlies pricing in derivatives, fixed income, and valuation more broadly.

Financial prices are not arbitrary; they are linked by replication and the absence of arbitrage. A quantitative analyst should therefore not interpret a model output in isolation. A model price that violates basic consistency conditions may signal an error in the model, a missing constraint, or an overlooked transaction cost, one reason financial modeling is both statistical and economic. The same logic extends to options, swaps, and structured products, where the challenge is often to identify the replicating portfolio and the correct discounting. Such relationships make finance a systematic discipline rather than a descriptive one: even when markets look noisy, strong forces keep valuations tied to economic logic.

A second example, a forward contract, makes the same point with a replicating portfolio rather than a single comparison bond. Suppose a stock trades today at $S_0 = \$100$, pays no dividends over the next year, and the risk-free rate is $r = 5\%$. A one-year forward contract obligates its holder to buy the stock in one year at a price fixed today; the no-arbitrage forward price is

$$F_0 = S_0(1 + r) = 100(1.05) = \$105,$$

the cost of replicating the forward's payoff by borrowing \$100 today at the risk-free rate, buying the stock now, and holding it for one year (a strategy known as cash-and-carry).

If the forward instead traded in the market at $F_{\text{mkt}} = \$110$, an arbitrageur could borrow \$100, buy the stock, and simultaneously sell (go short) the forward at \$110; in one year, the arbitrageur delivers the stock against the forward for \$110, repays the loan plus interest of $100(1.05) = \$105$, and keeps a riskless profit of $\$110 - \$105 = \$5$ wherever the stock price ends up, since the forward sale fixes the delivery price in advance. Chapter 5's option and derivative pricing methods generalize this relationship. An option's value, unlike a forward's, depends on the underlying's volatility because its payoff is not a fixed, symmetric obligation, but the same replication logic underlies both: construct a portfolio of tradable assets with an identical payoff, and price the derivative to match.

A Third Example: Triangular Arbitrage in Currency Markets

No-arbitrage pricing applies just as directly when three related prices must be mutually consistent, and foreign exchange markets provide the cleanest illustration. Three exchange rates, EUR/USD, GBP/USD, and EUR/GBP, are not independent: knowing any two implies the no-arbitrage value of the third, since converting EUR to USD and then USD to GBP must produce the same result as converting EUR to GBP directly. Suppose the market quotes EUR/USD = 1.10 (one euro buys \$1.10) and GBP/USD = 1.25 (one pound buys \$1.25), implying a no-arbitrage EUR/GBP rate of

$$\frac{\text{EUR/USD}}{\text{GBP/USD}} = \frac{1.10}{1.25} = 0.88,$$

one euro should be worth £0.88. If the EUR/GBP market instead directly quotes 0.87, euros are priced slightly cheaper against pounds directly than the USD route implies, and a triangular arbitrage exists: starting

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with £1,000,000,

$$\begin{aligned}\text{Step 1 (GBP} \to \text{EUR at 0.87):} &\quad 1{,}000{,}000/0.87 \approx €1{,}149{,}425.29,\\ \text{Step 2 (EUR} \to \text{USD at 1.10):} &\quad 1{,}149{,}425.29 \times 1.10 \approx \$1{,}264{,}367.82,\\ \text{Step 3 (USD} \to \text{GBP at 1/1.25):} &\quad 1{,}264{,}367.82 \times 0.80 \approx £1{,}011{,}494.25,\end{aligned}$$

a riskless profit of about £11,494, or roughly 1.15%, on the round trip, realized in seconds and requiring no view on whether any of the three currencies will rise or fall. As with the bond and forward, this mispricing cannot persist. High-frequency and algorithmic traders (Chapters 13 and 14) monitor such three-way consistency continuously, and their trading pushes the rates back into alignment within moments. Triangular arbitrage in liquid major-currency pairs is therefore too small and short-lived to exploit manually; today it is mainly a textbook illustration of a real economic force rather than a strategy open to anyone without a fully automated, low-latency system.

Frictions and the Limits of Arbitrage

Financial theory often assumes frictionless markets, but real markets are full of frictions: transaction costs, taxes, delays, information asymmetry, margin requirements, and limits on borrowing. These change the economics of investment and the behavior of prices.

If trading a security is expensive, an investor may need a larger expected return to hold it. If borrowing is restricted, a trader cannot always implement an otherwise profitable strategy. If information is not shared equally, some participants act faster or better informed than others. Even small frictions can have large aggregate effects, particularly when many participants trade at once.

Practitioners therefore distinguish theoretical from realized prices. Two assets with identical cash flows should in theory trade at the same value, but may not if short selling is difficult, settlement is delayed, or the market is temporarily illiquid. Such deviations create opportunities for some participants and risk for others, a tension between theory and practice that defines much of finance.

3.5 Fragility and Market Design

The same connections that make markets efficient also transmit failure. This part covers how stress propagates, how design and speed shape that propagation, and one afternoon in 2010 when it happened in minutes.

3.5.1 Financial Crises, Contagion, and Resilience

Understanding markets also requires understanding crises. Crises are not random accidents; they often emerge when leverage is high, liquidity is fragile, and institutions are highly interconnected. A seemingly small shock can then trigger forced selling, margin calls, and sudden changes in funding conditions, and a cascade of losses can spread across firms, asset classes, and countries. This is the pattern of the 2008 crisis discussed under clearing and systemic risk: mortgage defaults, amplified through securitization, derivatives, and short-term funding markets, became systemic because institutions that seemed healthy turned out not to be resilient once funding dried up.

For AI and quantitative finance, a model can be accurate in normal conditions and still fail in a crisis if the regime changes. The data may be nonstationary, relationships between variables may weaken, and strategies may become far more costly to execute, which is why model stress testing and scenario analysis are essential.

An earlier episode, the near-collapse of the hedge fund Long-Term Capital Management (LTCM) in 1998, teaches a related but distinct lesson: sophisticated quantitative models can fail because of leverage and crowding, not because the statistical reasoning was wrong. LTCM's strategies relied on historical relationships between related securities converging to normal levels, a reasonable premise most of the time, applied with very high leverage to turn small, reliable-looking spreads into large returns. When the 1997 Asian financial crisis and the 1998 Russian debt default triggered a broad flight to safety, those relationships moved sharply against LTCM. Because many other participants held similar positions, there was no way to exit without pushing prices further the wrong way. The near-failure was severe enough that the Federal Reserve organized

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a private-sector bailout to prevent broader disruption. The lesson is not that statistical models are unreliable, but that estimated relationships can break down when many participants rely on the same relationship and use leverage to size their bets on it, a crowding risk that a model built only from historical prices will rarely reveal.

3.5.2 Technology, Speed, and Market Design

Technology has transformed market design. Electronic trading, co-location, and automation have reduced costs and sped up order processing, improving access for many participants but also increasing complexity. The market now resembles a digital system in which algorithms interact at high speed and small delays can be very valuable.

Technology changes the structure of competition as well as the efficiency of trading. A participant with faster infrastructure may gain an edge without a better economic view, which is why market microstructure has become central to modern finance: price formation is shaped by trading rules, network effects, and technology as well as by fundamentals.

3.5.3 Case Vignette: The 2010 Flash Crash

On May 6, 2010, U.S. equity markets experienced one of the most dramatic intraday moves in modern history: major indices fell sharply within minutes, some stocks and exchange-traded funds briefly traded at absurd prices, pennies in some cases, and the market recovered most of its losses within the same session. The Flash Crash is a useful case study because the joint SEC-CFTC investigation identified a chain of mechanical, structural causes rather than any piece of adverse economic news.

The investigation traced the initial trigger to a large sell order in E-mini S&P 500 futures, executed by an automated algorithm designed to sell a fixed percentage of trading volume without regard to price or time. As this selling pressure pushed prices down, high-frequency trading firms, which had been providing a substantial share of market liquidity, began rapidly buying and reselling the same contracts to each other rather than absorbing the imbalance, a pattern sometimes described as "hot potato" trading.

Within minutes, many of these high-frequency firms and other liquidity providers withdrew entirely as their risk controls triggered, and depth (one of the liquidity dimensions of Section 3.4.1) evaporated on both the futures and the equity markets at once. With so little resting liquidity, ordinary sell orders walked prices down dramatically, the price-impact effect of Section 3.2.2 at a much larger and faster scale.

The Flash Crash illustrates several of this chapter's themes at once. Liquidity is not a guaranteed feature of a market but a behavior that participants can withdraw suddenly and together, especially under stress, the regime dependence of liquidity discussed in Section 3.6. Automated systems interacting with one another can produce dynamics, like the hot-potato pattern, that no individual system's designers intended or anticipated. And market fragmentation across linked venues and instruments has practical consequences: the crash began in a futures contract but spread almost instantly to equities, since closely related prices are tied together by the no-arbitrage forces of Section 3.4.3. In response, regulators introduced market-wide circuit breakers and individual stock "limit up-limit down" rules designed to pause trading automatically during extreme price moves, a structural safeguard that did not exist in May 2010.

3.5.4 Case Vignette: Archegos and Leverage Through Total Return Swaps

The Flash Crash showed liquidity vanishing in minutes. The collapse of Archegos Capital Management in March 2021 shows hidden leverage turning one stock's decline into more than $10 billion of losses at the banks that financed it. Archegos was the family office of Bill Hwang. According to the SEC, it grew from about $1.5 billion of capital with $10 billion of exposure in March 2020 to more than $36 billion of capital with $160 billion of exposure at its peak in March 2021, concentrated in a handful of stocks such as ViacomCBS (U.S. Securities and Exchange Commission 2022).

Most of that exposure was held through *total return swaps*, one of the swap variants mentioned earlier. In a total return swap the prime broker buys the stock as a hedge and pays the client the stock's full return, while the client pays a financing rate and posts margin. The client has the economics of owning the shares without owning them, and two consequences follow from the margin arithmetic of Section 3.4.2. First, leverage: an initial margin of m supports exposure of $1/m$ times the collateral, and a price fall of m wipes

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the collateral out. A cash purchase of stock is limited by the Federal Reserve's Regulation T to 50% initial margin, a leverage of 2× that survives a 50% fall. Swap margin is negotiated bilaterally, and Credit Suisse's own investigation found it held only about 10% against Archegos's swaps (Paul, Weiss, Rifkind, Wharton & Garrison LLP 2021), a leverage of 10× that survives a 10% fall. Second, invisibility: swap positions did not appear in public holdings filings, and each prime broker saw only its own book, so no one, the banks included, saw that Archegos's combined position in some stocks was a large fraction of their float.

The trigger was ordinary. On 22 March 2021 ViacomCBS closed at $100.34; after it announced a stock offering, the shares fell, Archegos could not meet its margin calls, and by the close on 26 March they stood at $48.23, a fall of 51.93%. On $10 billion of swap exposure at 10% margin that is a loss of $5,193.3 million against $1,000 million of collateral, leaving $4,193.3 million for the lenders to absorb. How it was shared depended on who sold first. Suppose, in a stylized version, that four prime brokers each hold an equal $2,500 million block, each with $250 million of collateral, and each block sold walks the price down another 12.5%, the order-book walk of Section 3.2.2 at the scale of billions. Each broker, considering only its own block, expects to sell at an average of 6.25% below the starting price, comfortably inside its 10% margin. But the four blocks hit the same order book in sequence. The first seller sells at an average 6.25% below the start and loses nothing; the second, at 18.75% below, loses $218.75 million beyond its collateral; the third, at 31.25%, loses $531.25 million; the fourth, at 43.75%, loses $843.75 million. Brokers lose $1,593.75 million in total, the fund loses $906.25 million of its $1,000 million of collateral, and the price ends 50% below where it started. This is the first-mover arithmetic of the stablecoin run in Worked Example 3.3, now among lenders: margin that looked adequate for each position alone was inadequate for all of them together.

The real losses were ordered much like that. Goldman Sachs, which sold large blocks early on Friday 26 March, was reported to expect an immaterial effect on its results, and Morgan Stanley, also an early seller, lost just under $1 billion. Credit Suisse, slower to act, lost about $5.5 billion and Nomura nearly $3 billion, and total bank losses exceeded $10 billion. Credit Suisse exited prime brokerage, and Hwang was convicted of fraud and market manipulation in July 2024 and sentenced to 18 years in prison in November 2024. For quantitative risk management the lessons are the ones LTCM taught a generation earlier: a margin model calibrated to one position's volatility ignores the liquidation impact of everyone else's identical position, and concentration that no single counterparty can see is concentration nonetheless.

3.6 Challenges in Modeling Market Structure

The institutional detail of this chapter creates specific, recurring challenges for anyone building data-driven or AI-based systems on financial markets.

Fragmented and opaque data. Much trading, especially in bonds, derivatives, and large institutional equity orders, occurs over-the-counter or in dark pools rather than on a single transparent exchange. A dataset built from public exchange feeds alone is an incomplete picture of the market, and models trained on it can miss where and how large trades occur.

Microstructure noise. At very short horizons, prices are affected by the mechanics of trading (bid-ask bounce, order-book dynamics, the strategic behavior of market makers) as well as by fundamental information. A high-frequency model that treats every price tick as a fundamental signal risks fitting this noise rather than anything economically meaningful, a problem revisited in the chapters on high-frequency trading.

Regime dependence of liquidity. Liquidity, spreads, and depth are not fixed properties of a security; they change with volatility, macroeconomic conditions, and stress, often when a trading or risk model most needs to work. A backtest run only over calm periods can substantially overstate a strategy's real-world performance.

Reflexivity and adversarial dynamics. Unlike a physical system, a market contains participants who adapt to observed patterns, including patterns generated by other algorithms. A signal or arbitrage relationship, once discovered and traded at scale, can weaken or disappear, so one-time model validation is not sufficient.

Modeling systemic effects. Asset-level or firm-level models rarely capture the network effects (interconnections between institutions, correlated leverage, shared counterparties) that drive systemic events such as the 2008 crisis. Modeling these system-wide dynamics remains much harder and less mature than single-asset prediction.

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Realistic transaction cost estimation. A strategy's profitability depends heavily on execution costs, market impact, and available liquidity, none fully known in advance and all easy to underestimate in a backtest that assumes trades execute at the observed price with no impact.

A purely statistical view of market data that ignores institutional structure is therefore unlikely to produce robust financial models, which is why the book treats market structure as foundational before turning to prediction and optimization.

3.7 Key Takeaways

Financial markets exist to allocate capital, transfer risk, and reveal information, and their institutions and instruments shape the opportunities and constraints investors and firms face. Debt, equity, and derivatives are the building blocks of modern finance, and how they are traded has direct consequences for liquidity, pricing, and risk management.

The order-book example in Section 3.2.2 showed that, with no news and no change in fundamental value, a large order can move the average execution price simply because liquidity is finite at each price level; the Flash Crash showed the same mechanism at market-wide scale when liquidity providers withdrew together. The swaps and regulatory material of this chapter (interest rate swaps, Dodd-Frank, MiFID II, Basel III) resurfaces in the risk management chapters, since modern risk management is defined as much by regulatory capital requirements as by statistical modeling choices. Payment rails and digital assets obey the same logic as securities settlement: the faster and more final a payment, the less can be undone afterward. A stablecoin promising redemption at par is only as sound as the duration and credit quality of its reserves, as the run arithmetic of Worked Example 3.3 shows. A central bank sets only the overnight rate; longer yields and stock prices move with what that rate reveals about the expected path ahead. The Archegos case showed the first-mover arithmetic of a stablecoin run playing out among the prime brokers that financed a single, invisibly concentrated fund. Market structure is not a fixed backdrop for trading; it shapes prices, costs, and the viability of any strategy built on it.

3.8 Suggested Exercises

3.1 Explain the difference between a bond and a stock in terms of cash flow rights and risk.

3.2 Describe the role of a market maker in a financial market.

3.3 Give one example of how a derivative can be used for hedging.

3.4 Explain why liquidity matters for asset prices.

3.5 Compare exchange-traded and over-the-counter markets in terms of transparency and counterparty risk.

3.6 Discuss one reason why a trading strategy might fail even if it has good historical predictive performance.

3.7 Explain how a bid-ask spread reflects the cost of immediacy in a market.

3.8 Choose one challenge from Section 3.6 and explain how it could cause a backtested trading strategy to look profitable historically but lose money in live trading.

3.9 Using the order book in Table 3.1, compute the average execution price and price impact (in dollars and basis points) for a market sell order of 300 shares, which will walk through the bid side of the book.

3.10 Explain, in your own words, the difference between the order-processing, inventory-holding, and adverse-selection components of the bid-ask spread, and give one market condition that would cause each component to widen.

3.11 Explain why an interest rate swap does not require the exchange of the notional principal amount, and describe a situation in which a firm would want to swap from floating-rate to fixed-rate interest payments.

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3.12 Compare the 2008 financial crisis and the LTCM episode described in this chapter: what role did leverage play in each, and how did the two episodes differ in their trigger and their transmission mechanism?

3.13 Using Section 3.4.2's method, a trader shorts 200 shares at \$80 with a 150% initial margin requirement and a 30% maintenance margin. Compute the total initial account equity and the share price at which a margin call is triggered.

3.14 Using Section 3.4.3's cash-and-carry logic, a non-dividend-paying stock trades at \$40 and the risk-free rate is 4%. Compute the no-arbitrage one-year forward price, and describe the specific trades an arbitrageur would execute today and in one year if the forward instead traded at \$43.

3.15 A dealer borrows \$25 million overnight in the repo market at an annualized rate of 4.5%, using the 360-day money-market convention from this chapter's repo example. Compute the dollar interest owed for one night and the total repurchase price.

3.16 Explain how the Flash Crash illustrates the "regime dependence of liquidity" challenge from Section 3.6, using specific details from the case study.

3.17 Using this chapter's interest rate swap example, recompute the swap's value to the fixed-rate payer if the one-year and two-year discount rates were instead 5.5% and 6% (rates have risen above the 5% fixed rate rather than fallen below it), and explain in words why the sign of the swap's value to the fixed-rate payer flips relative to the original example.

3.18 Using this chapter's triangular arbitrage example, suppose the EUR/GBP rate is instead quoted at 0.89 rather than 0.87 (EUR/USD and GBP/USD unchanged). Determine whether the arbitrage cycle described in the text ($£ \to € \to \$ \to £$) is still profitable, and if not, describe which direction around the triangle would now be profitable instead.

3.19 Rework Worked Example 3.3's stablecoin run with MiCA's minimum cash share: the issuer holds $C = \$3{,}000$ million of deposits and $B = \$7{,}000$ million of zero-coupon Treasuries bought at 4%, and yields rise by 200 basis points. For three-month bills and for two-year zeros, use Equation (3.3) to compute the mark-to-market loss, then use Equation (3.4) to compute reserves per remaining token after 30% and after 60% of tokens are redeemed. How much of each redemption must be met by selling Treasuries, and how does the larger cash buffer affect the realized loss compared with the reserves per remaining token?

3.20 Using Table 3.2, explain why an authorized push payment scam is harder to reverse on an instant payment rail than an unauthorized card transaction is on a card network, and what this implies about when a fraud model must make its decision. Then name one thing that tokenizing a money-market fund changes about how its shares settle and one risk of holding the fund that tokenization leaves unchanged.

3.21 Rework Section 3.1.5's rate-transmission example for a 25-basis-point cut that the market expects to persist, so that the expected one-year rates for the next two years are both 4.75%. Compute the new two-year yield and the percentage change in the price of the two-year zero (starting from \$90.7029), and the percentage change in the Gordon-growth stock price if its discount rate falls from 8% to 7.75% ($D_1 = \$4$, $g = 4\%$). Why does the stock move so much more than the bond?

3.22 Redo Section 3.5.4's stylized fire-sale race with a swap margin of 20% instead of 10% (four brokers, \$2,500 million blocks, 12.5% price impact per block). What leverage does the fund now have, which brokers lose money beyond their collateral, and what are total broker losses? Compare the answer with a long position bought in cash under Regulation T's 50% initial margin.

3.23 Using Section 3.1.6, classify each of the following as buy side, sell side, or neither, and state how each earns its revenue: a pension fund, an investment bank's equity derivatives desk, a credit rating agency, a quantitative hedge fund, and a prime broker. Then explain the issuer-pays conflict of interest, and why it mattered more for mortgage-backed securities before 2008 than for large corporate bonds.

3.24 **Challenge (real data).** Estimate Roll's implied spread, Equation (3.2), from daily closing prices. Download unadjusted daily closes for 3 January to 29 December 2023 for the S&P 500 ETF SPY and for the small-cap furniture maker Flexsteel Industries (FLXS) with `yfinance` (`auto_adjust=False`). (a) For each, compute the first-order autocovariance of daily price changes over the full year and,

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where it is negative, the Roll spread in dollars and in basis points of the average price. (b) Repeat month by month, and count the months in which the estimate is undefined because the autocovariance is positive. (c) SPY's quoted spread is about one cent. Explain what your daily-data estimates for SPY say about the estimator's reliability for liquid securities at daily frequency, and why it may be more informative for an illiquid small-cap.

3.9 Further Reading

The following books treat these topics in more depth:

- Bodie et al. (2024), *Investments*. A broad introduction to markets, instruments, and institutions that complements this chapter's overview with a fuller treatment of portfolio and asset-pricing theory.
- Hull (2022), *Options, Futures, and Other Derivatives*. The standard reference for the derivatives instruments introduced briefly in this chapter, developing their pricing and hedging in full in Chapter 5.
- Fabozzi (2007), *Fixed Income Analysis*. Goes well beyond this chapter's brief bond-market coverage into the pricing, risk, and institutional detail of fixed-income securities.
- Shleifer (2000), *Inefficient Markets*. A behavioral-finance counterpoint to the efficient-market and price-discovery discussion in this chapter, examining the frictions and limits to arbitrage that keep prices from always reflecting fundamentals.
- O'Hara (1995), *Market Microstructure Theory*. The foundational academic treatment of the order-book mechanics, bid-ask spread decomposition, and market-making economics introduced in this chapter's worked examples.
- Gorton and Zhang (2023), "Taming Wildcat Stablecoins." Argues, from the history of nineteenth-century U.S. private banknotes, that a stablecoin redeemable at par is a form of bank deposit exposed to runs, the logic behind this chapter's stablecoin worked example and the full-reserve rules that followed.
- Financial Crisis Inquiry Commission (2011), *The Financial Crisis Inquiry Report: Final Report of the National Commission on the Causes of the Financial and Economic Crisis in the United States*. The official U.S. inquiry into the 2008 crisis, including its case study of how issuer-paid ratings were assigned to mortgage securities, the background to this chapter's discussion of rating agencies.
- Bank for International Settlements (2025), "The Next-Generation Monetary and Financial System." The Bank for International Settlements' assessment of stablecoins against the tests of sound money, and its case for tokenized central bank reserves, deposits, and government bonds on a shared ledger.

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